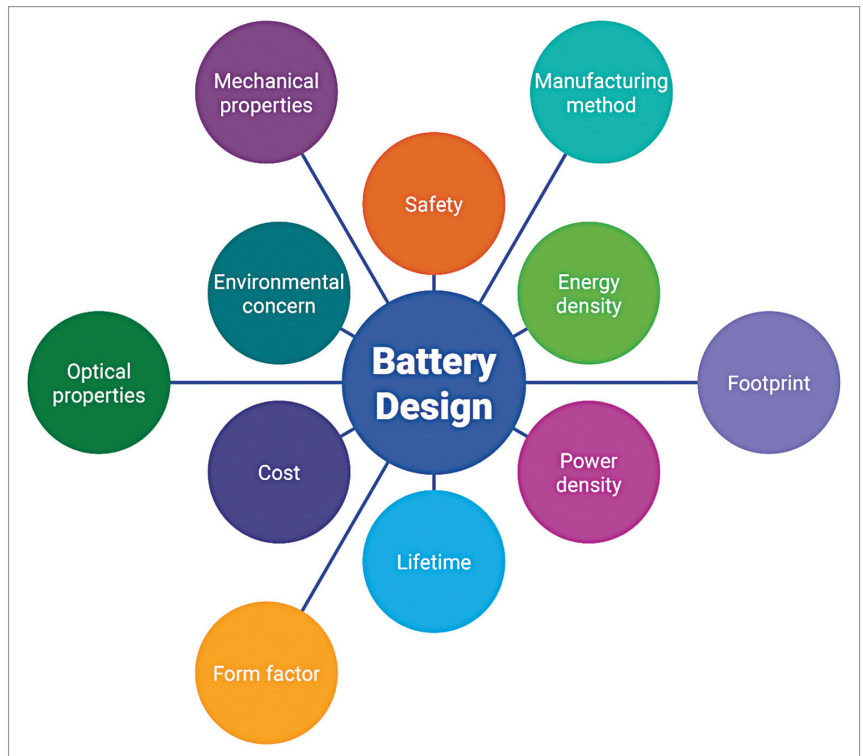


polymer matrix embedded with a gel, dried to a semi-solid state to balance flexibility with better energy capacity, though still below that of liquid electrolytes.

Despite their high energy density, liquid electrolytes present significant risks in flexible applications due to potential leakage and combustion when bent. To mitigate these risks, manufacturers have developed advanced packaging solutions, like pre-crumpled designs to control the direction of stress to prevent component puncture. However, such batteries often have a higher bending radius, reflecting reduced flexibility.

The bending radius, a crucial measure for flexible batteries, indicates the minimum radius a battery can achieve when bent. A smaller radius indicates higher flexibility, which is suitable for tightly bending applications.

Solid-state and semi-solid electrolytes are generally safer than liquid options, being less prone to leakage and air reactions. Yet, solid-state electrolyte technology, being relatively new, lacks the extensive testing that liquid electrolytes have undergone. Ongoing development and testing are vital to ensure that flexible batteries can be safely integrated into compact, bendable systems.



Battery properties and their importance (Source: IDTechEx)

Leading the way in green innovation

Flexible battery manufacturing is dynamically evolving, leveraging existing infrastructure to enhance production without the need for comprehensive overhauls. The established use of pouch cells

exemplifies this trend, serving as a bedrock for innovation and allowing for the integration of more standardised printing technologies. These advancements have been crucial in enabling industrial printers to efficiently produce a wide array of electronic components, thereby scaling up the production capabilities of flexible batteries.

As environmental concerns become increasingly paramount, the industry is shifting towards more sustainable manufacturing practices. Specifically, within the thin battery space, there is a growing preference for zinc-based batteries over traditional lithium-based alternatives, primarily due to zinc's greater abundance and lower carbon emissions during extraction. Leading this sustainable change, research institutes like Germany's Fraunhofer are pioneering the development of batteries that not only reduce operational times and chemical use

Regulatory and safety standards overview

Market regulations: North America vs Europe

- **Stricter standards:** Europe typically enforces stricter regulations compared to North America, yet both regions uphold more stringent rules for medical devices, including implants and wearables, than for non-medical devices.

Flexibility and safety testing

- **Rigorous testing:** Battery manufacturers are required to prove that their products can repeatedly bend and curve without significant safety risks. This involves extensive testing, often including over 1000 flexing cycles, where the only acceptable result is minimal battery capacity loss.

Implanted medical devices

- **Material and packaging safety:** For devices implanted in the body, the regulatory standards are exceptionally high. Materials must not negatively interact with blood, and packaging must be secure enough to prevent any leakage, especially important when using semi-solid or liquid electrolytes. These measures ensure device safety and reliability, which are essential in critical healthcare applications.